

West Hartford Deck — Lumber Yard Purchase List

Preparer: Hollis | **Date:** 2026-05-11 | **Rev:** v18 / 2026-05-11

Scope: Attached 23'-6" × 16'-6" deck, pressure-treated framing on Pylex screw piers, MoistureShield Vision composite decking. This list covers all lumber-yard and specialty-supply items required for framing, decking, connectors, fasteners, and moisture protection. Pylex 50 screw piers are ordered separately — not a lumber-yard item.

1. Shims & Bearing Hardware

Item	Spec	Qty	Notes
¼" cedar or HDPE rot-resistant shims	Min. 1.5" × 3.5" × ¼" shim stock	40	2 shims per saddle × 2 sides × 10 saddles = 40 total. Use cedar or HDPE only — no OSB, no wood composite, no plywood.
½" HDG lag bolt (girder-to-saddle)	½" × 3" HDG	20	2 per saddle × 10 saddles, up into girder bottom.

2. Lumber — Pressure-Treated SYP, Ground-Contact (UC4B)

All PT lumber: Southern Yellow Pine, ground-contact rated (UC4B), KDAT (kiln-dried after treatment) preferred. Verify net dressed length at the yard before delivery.

Category	Stock Size	Qty to Order
Ledger	2×10 PT 20-ft	1
Girder plies	2×10 PT 12-ft	7 (6 needed + 1 spare)
Girder plies	2×10 PT 8-ft	5 (4 needed + 1 spare)
Joists	2×10 PT 8-ft	55 (50 needed + ~10% waste)
		—

Category	Stock Size	Qty to Order
Blocking (field + house-side)	2×10 PT cuts from drops	

Girder splice recipe: Each girder line is a doubled 2-ply 2×10. Ply 1 uses two 11'-6" pieces cut from 12-ft stock (6" waste each); Ply 2 end pieces are 5'-9" cut from 8-ft stock, with an 11'-6" center section also from 12-ft stock. Splices are staggered between plies per standard practice. Order 4 extra 8-ft boards if dedicated blocking stock is preferred over using joist drop-offs.

3. Decking, Picture Frame & Fascia

All MoistureShield Vision composite, capped, DiamondDefense scratch coat, 50-yr fade & stain warranty. Available at Home Depot, Lowe's, Richards Building Supply (Hartford metro).

Product	Length	Qty to Order
MoistureShield Vision field planks	12-ft	72
MoistureShield Vision center divider breaker	16-ft or 20-ft (1 piece)	1
MoistureShield Vision PF boards	24-ft (house-side + far-side)	2
MoistureShield Vision PF boards	18-ft (left-side + right-side, 16'-3")	2
MoistureShield Vision fascia	24-ft (far-side)	1
MoistureShield Vision fascia	16-ft (left-side + right-side — exact length)	2

4. Connectors & Hangers

All connectors: hot-dip galvanized (HDG), ACQ-compatible. Simpson Strong-Tie specified throughout.

Item	Model	Qty	Notes
Girder end-to-ledger hanger (doubled 2×10)	Simpson HU210-2 (2-ply, HDG)	2	One at each end of each girder line at the ledger (x=0' and x=23'-0') — only the 2 end points of the mid-girder line land at the ledger;

Item	Model	Qty	Notes
Joist-to-ledger face-mount hanger	Simpson LUS210Z HDG	25	outer-rim girder does not connect to ledger. 25 joist rows × 1 hanger each at ledger end
Joist-to-mid-girder face-mount hanger (house-side face)	Simpson LUS210Z HDG	25	25 joist rows × 1 hanger each hanging from mid-girder house-side face
Joist-to-mid-girder face-mount hanger (far-side face)	Simpson LUS210Z HDG	25	25 joist rows × 1 hanger each hanging from mid-girder far face
Joist-to-outer-rim-girder face-mount hanger (inboard face only)	Simpson LUS210Z HDG	25	25 joist rows × 1 hanger each; joists terminate at outer-rim girder inboard face

Hanger count total — LUS210Z: 25 (ledger) + 25 (mid-girder house face) + 25 (mid-girder far face) + 25 (outer-rim girder) = **100 LUS210Z**. Add 5% overage: **order 106 LUS210Z** (round up to 110 for convenience).

5. Fasteners

All fasteners: HDG or stainless (ACQ-compatible). No electroplated zinc.

5a. Ledger-to-House

Item	Spec	Qty	Notes
Ledger lag bolts (field, between hanger locations)	½" × 4" HDG lag	34	½" HDG lag at 18" o.c. two-row staggered between hanger-connection zones.
Through-bolts at girder-to-ledger (2 locations)	½" × 6" HDG carriage bolt + nut + 2 washers	12	6 per cluster × 2 cluster locations (x=0', x=23'-0') at mid-girder line end-to-ledger connections.
Ledger through-bolt backup blocks (interior band joist)	2×10 PT scrap	2 short pieces	Back-blocking behind ledger at each carriage-bolt cluster.

5b. Girder Assembly (Doubled 2-Ply)

Per AWC DCA-6 / NDS, the specified method for laminating a 2-ply 2×10 girder is ½” through-bolts staggered along the full run. **Spec: ½” × 3½” HDG carriage bolt + nut + 2 washers at 24” o.c., staggered top-and-bottom, full girder run both lines.**

Item	Spec	Qty	Notes
Ply-to-ply carriage bolts + hardware (AWC DCA-6)	½” × 3½” HDG carriage bolt + nut + 2 washers	50	~1 per 24” o.c. × 46’ total girder run (2 lines × 23’) = ~23 per line × 2 lines = ~46; round to 50.

5c. Joist Hangers

Item	Spec	Qty	Notes
Joist hanger nails/screws — LUS210Z	Simpson N10 (10d × 1½”) or SDS ¼”×1½” screws per Simpson spec sheet	~1,400	100 hangers × ~14 nail holes per LUS210Z = 1,400; buy 1,500.
HU210-2 fasteners (girder-end-to-ledger)	Per Simpson spec (SDS ¼”×1½” or N10 as spec sheet requires for 2-ply)	~60	2 hangers × ~30 fastener holes each.

Critical: Verify exact nail/screw specification on the Simpson packaging for each hanger model. Do NOT substitute nail sizes.

5d. Decking Fasteners

Item	Brand / Spec	Qty	Notes
Hidden fasteners — field planks	Tiger Claw TC-G clips	~700	25 joist rows × 32 planks × ~16 clip points per plank = 512 clips per half × 2 halves; buy 700 with overage.
#8 stainless screws for TC-G clips	Tiger Claw specified (stainless, 1-5/8”)	~700	One per clip.
Cortex face-screws + color-matched	Cortex for MoistureShield (must match Vision colorway)	5 boxes (100-ct each)	~4 screws per lineal foot × ~80 lf four-sided PF + 16 lf

Item	Brand / Spec	Qty	Notes
plugs — PF boards (4 sides + center divider)			center divider = ~384 screws.
Stainless trim screws — fascia to rim/drop-board	#10 × 2½” stainless	1 box (100-ct)	~3 screws per lineal foot × ~55 lf of fascia ≈ 165.

5e. Lateral Tie Fasteners

Item	Spec	Qty	Notes
DTT2Z structural screws	SDS ¼”×4½” per Simpson spec	~ 32	~8 per DTT2Z device × 4 devices.

6. Flashing & Moisture Protection

Item	Brand / Spec	Coverage / Size	Qty	Notes
Peel-and-stick flashing tape (ledger)	Henry Blueskin VP160 or Carlisle CCW-705	6” wide roll	2 rolls (75 lf each)	Apply to house sheathing behind ledger before ledger install; 23’-0” run.
Z-flashing, aluminum	2” leg × 2” leg, 10-ft sticks	3 sticks		Tucked behind siding/housewrap above top of ledger.
Kick-out flashing	Aluminum or plastic, pre-bent	2		At each end of ledger where siding terminates.
Joist / girder top tape	Vycor Deck Protector 2” or Trex Protect 2½”	2” wide roll, 50 lf per roll	10 rolls	Apply to top of every joist segment (50 segments × ~8 lf each ≈ 400 lf), both

Item	Brand / Spec	Coverage / Size	Qty	Notes
				girder top runs (~46 lf each × 2 ≈ 92 lf), and ledger top (23 lf) = ~515 lf total. Buy 11 rolls to be safe.
End-grain sealant	Copper Green Wood Preservative or Anchorseal 2	1-qt brush-on	2 qt	Seal ALL PT field cuts immediately after cutting.
Backer rod + polyurethane sealant	Sikaflex 1A or similar, gray	1 tube	3 tubes	Ledger-to-house seam caulk at ledger penetrations.
Ledger spacers / standoffs	Starborn Camo LedgerUp composite or PT shims	—	as needed	If siding extends behind ledger location.
Drip cap (head flashing)	Aluminum drip cap, 2" face	24-lf sticks	1 stick	Over ledger at top if housewrap alone is insufficient.

7. Lateral Tie System (IRC R507.9)

Item	Model	Qty	Notes
Deck-to-house lateral connector	Simpson DTT2Z (or MTWDeck equivalent)	4	IRC R507.9 minimum; deck is 16'-3" wide. Each device connects deck framing through ledger to interior floor framing. Basement accessible.

Placement: space DTT2Z devices evenly along ledger, avoiding the HU210-2 hanger cluster zones. Confirm exact placement with structural engineer or West Hartford building official.

Scope Notes

Far-edge step ribbon removed intentionally. The far-edge step (formerly Section 5e) sits on precast concrete blocks — its foundation is out of scope for a lumber-yard purchase list and has been omitted from this document. Order the step's ribbon lumber and hardware separately when the concrete-block foundation is addressed.

Pylex 50 screw piers are ordered separately — not a lumber-yard item. Shims and girder-to-saddle hardware (Section 1) are included here as they ship with or alongside lumber-yard delivery.